

UNIT 1

COMPUTER LANGUAGES



Learning Objectives

- Understand the concept of computer languages.
- Identify different types of computer languages.
- Learn about high-level and low-level languages.
- Understand the importance of programming languages in computer operations.
- Recognize popular programming languages used today.

Introduction to Computer Language

A computer language is a way through which humans can communicate with computers and instruct them to perform specific tasks. Just like we use languages such as English, Urdu, and others to communicate with people, computers need a language to understand the instructions given to them. These languages allow us to control and interact with the computer effectively.

There are various types of computer languages, each serving a specific purpose. They can be broadly classified into two categories: **high-level languages** and **low-level languages**.

Types of Computer Languages

1. Low-Level Languages

- Low-level languages are the closest to machine language and are harder for humans to understand.
- **Machine Language:** This is the language that the computer understands directly. It uses binary code (0s and 1s) to represent instructions.
- **Assembly Language:** A slightly higher level than machine language, assembly uses symbolic codes and mnemonics, making it a bit easier for humans to write and understand. However, it is still very complex.



2. High-Level Languages

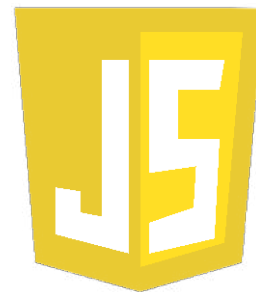
High-level languages are easier for humans to understand and write. They are closer to human languages and are used for creating applications and programs.

Some examples of high-level languages include:

- **Python** A widely-used, simple-to-learn programming language.
- **Java**: A versatile language used for web applications, Android apps, and more.
- **C++** Used for creating computer software, games, and other applications.
- **JavaScript** Commonly used for web development to create interactive websites.



Python



JavaScript:



C++



Java

Why Do We Need Computer Languages?

Computers operate in a way that is different from how humans think. While humans use spoken or written languages to communicate, computers require a different set of instructions to function. These instructions are written in programming languages.

- **Communication:** Computer languages allow us to communicate with computers, telling them what to do.
- **Automation:** With the help of computer languages, tasks such as calculations, data processing, and internet browsing are automated.
- **Application Development:** Computer languages allow developers to create software programs, websites, games, and applications.

Key Characteristics of Computer Languages

1. **Syntax:** Every programming language has a specific set of rules, called syntax, that determines how the instructions should be written. Just like in grammar, there are rules in programming languages to ensure the code is understood by the computer.
2. **Semantics:** This refers to the meaning behind the syntax. It defines how the instructions will behave when the program is run.
3. **Efficiency:** Some computer languages are designed to be fast and efficient, while others are designed for ease of use. For example, Python is easy to learn but may not be as fast as C++.

Basic Concepts of Programming

Programming is the process of writing instructions for a computer to follow. Here are a few basic programming concepts that are common across many languages:

- **Variables:** A variable is a container used to store data that can change. For example, you can store your age in a variable called "age."
- **Loops:** Loops allow you to repeat a block of code multiple times. For example, you can use a loop to display a message 10 times.
- **Conditional Statements:** These allow the program to make decisions. For example, an if statement checks if a condition is true and then runs certain code based on that condition.

Popular Programming Languages

1. Python

Python is known for its simplicity and readability. It is widely used for developing websites, software, and even artificial intelligence applications.

2. Java

Java is an object-oriented programming language that can be used to create mobile apps, web apps, and games. It is used by large organizations and is known for its portability.

3. C++

C++ is a powerful language used in system programming, game development, and creating large-scale applications.

4. JavaScript

JavaScript is the main language used to build interactive websites. It runs in web browsers, making websites dynamic and interactive.

5. Scratch

Scratch is a block-based visual programming language designed for beginners, especially kids. It allows students to create simple games and animations by dragging and dropping blocks instead of writing code.

Did You know?

The first computer programming language, **Fortran**, was created in 1957 and is still used in scientific applications today!

EXERCISES

01: Choose the correct answer.

1. Which of the following is a high-level programming language?

- Assembly Language
- Python
- Machine Language

2. Which language is used to develop websites?

- C++
- JavaScript
- Python

3. Which programming language is commonly used for artificial intelligence?

- Java
- Python
- JavaScript

4. What does "syntax" mean in computer programming?

- The logical meaning behind the code
- The set of rules for writing code correctly
- The speed of the code execution

5. Which language is considered a low-level language?

- Python
- Assembly Language
- Java

6. What is a key feature of Python programming language?

- It is difficult to learn
- It uses block-based coding
- It is simple and easy to learn

7. What type of code does a computer understand directly?

- Assembly Code
- Machine Code
- High-Level Code

8. Which programming concept allows repeating a block of code multiple times?

- Conditional Statements
- Loops
- Variables

9. Which of the following is NOT a high-level language?

- Java
- C++
- Assembly Language

10. What does "semantics" refer to in programming?

- The set of rules for writing code
- The meaning behind the code's syntax
- The structure of programming languages

02: Fill in the blanks.

1. The language used to write code that the computer directly understands is called _____.
2. _____ is a high-level programming language known for its ease of use and is commonly used for web development.
3. _____ is a low-level language that uses binary code to communicate with the computer.
4. _____ is a type of computer language that is easy for humans to read and write.
5. The programming language used for developing interactive websites is _____.
6. In computer programming, the rules for writing code are called _____.

03: Tick True or False for each statement.

	TRUE	FALSE
Assembly language is easier to understand than machine language.	☐	☐
Python is a low-level programming language.	☐	☐
Java is primarily used to develop websites.	☐	☐
JavaScript is used for web development and makes websites interactive.	☐	☐
The syntax of a programming language determines how the code should be written.	☐	☐
C++ is a popular high-level programming language used for creating mobile applications.	☐	☐

04: Match the terms with their correct definitions.

Column A	Column B
Java Script	A language used for web development
Python	Close to machine language, harder for humans to understand
Low-level languages	A programming language known for simplicity and readability
Assembly Language	Bit easier than machine language for humans to understand

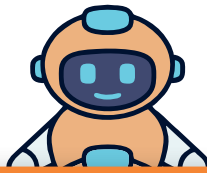
05: Answer the following questions briefly.

1. What is a computer language ?
2. What are the two main categories of computer languages?
Briefly explain each.
3. Name three examples of high-level programming languages and mention their uses.
4. Explain what a variable is in programming and give an example.
5. Why is Python considered a beginner-friendly programming language?

6: Provide detailed answers for the following questions.

1. Explain the difference between high-level and low-level programming language. Provide examples of each and their advantages.
2. Describe the process of creating a program using a high-level language like Python or Scratch. Include the key steps involved, such as defining variables and using loops.

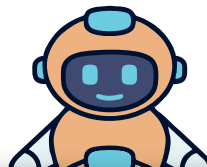
Lab Activity :1



Compare Low-Level and High-Level Languages:

Research and list the key differences between machine language, assembly language, and a high-level language like Python. Write a short report.

Lab Activity: 2



Explore Scratch Programming:

Create a simple project using Scratch. Use basic blocks like loops and conditional statements to make an interactive animation.